

# A Generalized Hole-Filling Strategy for Coprime Array Design

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**Abstract**—Sparse arrays have received great attention in direction-of-arrival (DOA) estimation due to their potential to increase degrees of freedom (DOFs) and reduce mutual coupling (MC). Previous works have devised several hole-filling strategies for specific prototype arrays, while a general framework for optimal sensor augmentation remains underexplored. In this paper, we propose a generalized hole-filling strategy based on coprime-type geometries to significantly increase uniform DOFs (uDOFs) with minimum extra sensors introduced. We formulate an optimization problem to derive the parameters, including interelement spacings and the number of sensors in subarrays, that maximize uDOFs under a given sensor constraint. Simulation results validate the superiority of the proposed array in terms of uDOFs while maintaining favorable MC suppression. Consequently, the proposed structure exhibits improved accuracy in DOA estimation under different MC conditions.

**Index Terms**—Sparse array, hole-filling strategy, uDOFs, mutual coupling leakage, DOA estimation.

## I. INTRODUCTION

Due to the simple and regular geometry, uniform linear arrays (ULAs) have been widely adopted in direction-of-arrival (DOA) estimation in various applications [1]–[3]. However, ULAs are limited in the maximum number of identifiable sources, which must be fewer than the number of physical sensors [4]–[6]. The interelement spacing of ULAs is typically constrained to half a wavelength to avoid spatial aliasing. Such dense configurations inevitably lead to severe mutual coupling (MC), which can degrade estimation performance if not properly mitigated [7]–[9]. Recent researches on sparse arrays offer promising alternatives to these issues. Specifically, leveraging the difference coarray (DCA) constructed from the pairwise sensor positions, a  $T$ -sensor sparse array can provide  $\mathcal{O}(T^2)$  degrees of freedom (DOFs), thus allowing for an underdetermined DOA estimation. Besides, the dilated interelement spacings in sparse arrays contribute to reduced MC, further enhancing their robustness in practical scenarios.

Nested-type geometries [10]–[13] were proposed by cascading multiple ULAs with different interelement spacings, offering closed-form expressions for sensor positions and available DOFs. Notably, they are able to generate a hole-free DCA, thereby providing a large number of uniform DOFs (uDOFs), whereas the compactly assembled subarrays may lead to severe MC. The coprime-type geometries [14]–[16] were constructed by interleaving a pair of sparse ULAs whose sensor counts and interelement spacings are coprime. While this configuration reduces MC, it inevitably introduces holes in the DCA, resulting in limited uDOFs. Two generalized coprime arrays [17], namely the coprime array with compressed interelement spacing (CACIS) and the coprime array with displaced subarrays (CADiS),

were devised by shrinking the interelement spacing and displacing subarrays. However, CACIS suffers from high redundancy between the self-DCA and cross-DCA and CADiS exhibits a fragmented DCA structure. The limitations have motivated further innovations, such as the thinned coprime array (TCA) [18] and the relocating extended coprime array (RECA).

In addition, several hole-filling strategies [19]–[21] have been proposed to enhance uDOFs of coprime-type arrays while remaining susceptible to MC effects and lacking generality beyond their specific prototype array configurations. In our previous work, we investigated common features of sparse structures and introduced a generic-coarray concept, characterized by element spacings and variable ranges, to develop an enhanced and generalized coprime array (EGCA) [22]. Although the EGCA exhibits enhanced uDOFs and great robustness to MC, it still lacks the generality to be extended to universal array structures.

In this paper, we propose a generalized hole-filling strategy (GHOST) for coprime-type prototype arrays from the generic-coarray viewpoint. As the generic-coarray covers several DCAs of multiple array configurations, the GHOST can be applied to diverse prototype arrays, thereby adapting to a wide range of scenarios. A rigorous optimization algorithm is provided with a tunable parameter to maximize uDOFs for a given number of sensors. Simulation results demonstrate the superiority of the proposed array configured by GHOST in terms of uDOFs and DOA estimation accuracy.

## II. COARRAY SIGNAL MODEL

Assuming  $K$  far-field, narrow-band, and uncorrelated signals from azimuths  $\{\theta_k\}_{k=1}^K$  are observed by a  $T$ -element sparse array with sensors located at  $\mathcal{P} = \{p_i\}_{i=1}^T$  (note that we omit the unit interelement spacing, which is half of the carrier wavelength, for brevity). Incorporating the MC matrix, the  $l$ -th observation is described as

$$\mathbf{x}(l) = \mathbf{C}\mathbf{A}\mathbf{s}(l) + \mathbf{n}(l), \quad (1)$$

where  $\mathbf{A} = [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \dots, \mathbf{a}(\theta_K)]$  is the array manifold with  $\mathbf{a}(\theta_k) = [e^{-j\pi p_1 \sin \theta_k}, \dots, e^{-j\pi p_T \sin \theta_k}]^T$  denoting the steering vector for the  $k$ -th signal;  $\mathbf{s}(l) = [s_1(l), s_2(l), \dots, s_K(l)]^T$  represents the signal vector and  $\mathbf{n}(l) = [n_1(l), n_2(l), \dots, n_K(l)]^T$  is the Gaussian white noise vector independent of  $\mathbf{s}(l)$ ;  $\mathbf{C}$  implies the MC matrix with  $B$ -banded symmetric form [23], whose  $(i, j)$ -th element is given as

$$\mathbf{C}(i, j) = \begin{cases} c_{|p_i - p_j|}, & |p_i - p_j| \leq B \\ 0, & \text{otherwise} \end{cases}, \quad (2)$$

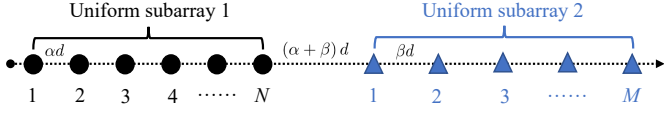


Fig. 1. Generalized CADiS.

where  $B$  denotes the coupling cut-off distance,  $c$  is the self-coupling coefficient,  $c_1 = ce^{j\pi/3}$  and  $\{c_m = c_1 e^{-j(m-1)\pi/8}/m\}_{m=2}^B$ . The coupling leakage of a sparse array is calculated as [22]

$$\mathcal{L} = \frac{\|\mathbf{C} - \text{diag}(\mathbf{C})\|_F}{\|\mathbf{C}\|_F}. \quad (3)$$

The covariance matrix of  $\mathbf{x}(l)$  is represented as

$$\mathbf{R} = \mathbb{E}[\mathbf{x}(l)\mathbf{x}^H(l)] = \mathbf{C}\mathbf{A}\mathbf{R}_s\mathbf{A}^H\mathbf{C}^H + \sigma_n^2\mathbf{I}_T, \quad (4)$$

where  $\mathbb{E}[\cdot]$  denotes the statistical expectation,  $\mathbf{R}_s$  is the source covariance matrix, and  $\sigma_n^2$  denotes the noise power. In the case of finite snapshots,  $\mathbf{R}$  is estimated as

$$\hat{\mathbf{R}} \approx \frac{1}{L} \sum_{l=1}^L \mathbf{x}(l)\mathbf{x}^H(l), \quad (5)$$

where  $L$  is the number of snapshots.

The vectorized form of  $\mathbf{R}$  can be obtained as

$$\mathbf{r} = \text{vec}(\mathbf{R}) = (\mathbf{C}^* \otimes \mathbf{C})(\mathbf{A}^* \circ \mathbf{A})\mathbf{p} + \sigma_n^2\mathbf{i}, \quad (6)$$

where  $\text{vec}(\cdot)$  is the vectorization operator,  $\otimes$  and  $\circ$  denote the Kronecker and Khatri-Rao products;  $\mathbf{p} = [\sigma_1^2, \dots, \sigma_K^2]^T$  denotes the signal power vector and  $\mathbf{i} = \text{vec}(\mathbf{I}_T) = [\mathbf{e}_1^T, \dots, \mathbf{e}_T^T]^T$ . Subsequently,  $\mathbf{G} = (\mathbf{A}^* \circ \mathbf{A})$  can be regarded as the manifold of the virtual array, whose sensor positions can be given by DCA

$$\mathcal{G} = \{p_i - p_j \mid p_i, p_j \in \mathcal{P}\}. \quad (7)$$

### III. GENERALIZED HOLE-FILLING STRATEGY

The generic-coarray [22] with coprime factors  $\alpha, \beta$  is defined as

$$\mathcal{G} = \{g \mid g = \alpha n + \beta m, n \in \langle N_1, N_2 \rangle, m \in \langle M_1, M_2 \rangle\},$$

where  $M_i, N_i \in \mathbb{Z}, i = 1, 2$  and  $\langle a, b \rangle$  represents the set of integers from  $a$  to  $b$  inclusive.

It has been proved that when  $\alpha \in \langle 1, M_2 - M_1 + 1 \rangle, \beta \in \langle 1, N_2 - N_1 + 1 \rangle$ , the range of uDOFs for  $\mathcal{G}$  satisfies

$$\mathcal{U}_c = \langle U_1 + U_0, U_2 - U_0 \rangle, \quad (8)$$

where  $U_i = \alpha N_i + \beta M_i, i = 1, 2$  and  $U_0 = (\alpha - 1)(\beta - 1)$ .

Without loss of generality, we propose the generalized hole-filling strategy (GHOST) based on a generalized CADiS whose DCA is a generic-coarray. As shown in Fig.1, the generalized CADiS consists of two subarrays with  $M, N$  sensors whose interelement spacings are  $\alpha d, \beta d$ , respectively. The displacement between two subarrays is  $(\alpha + \beta)d$ . To be specific,  $d$  is the unit spacing and  $\alpha, \beta$  are coprime. The sensor positions are expressed as

$$\begin{aligned} \mathcal{P} &= \mathcal{P}_1 \cup \mathcal{P}_2, \\ \mathcal{P}_1 &= \{\alpha n, n \in \langle 0, N-1 \rangle\}, \\ \mathcal{P}_2 &= \{\alpha N + \beta m, m \in \langle 1, M \rangle\}, \end{aligned} \quad (9)$$

where  $\mathcal{P}_1$  and  $\mathcal{P}_2$  denote the locations of the first and second subarrays, respectively. Defining  $\mathcal{G}_1$  as the DCA of  $\mathcal{P}$ , it can be represented as

$$\begin{aligned} \mathcal{G}_1 &= \{g_1 \mid g_1 = \alpha n + \beta m, n \in \langle 0, N \rangle, m \in \langle 0, M \rangle\} \\ &\quad \setminus \{\alpha N, \beta M\}. \end{aligned} \quad (10)$$

where  $\mathcal{A} \setminus \mathcal{B}$  denotes the difference set of  $\mathcal{A}$  and  $\mathcal{B}$ . Then according to [22], we can obtain the following theorem.

*Theorem 1:* Suppose the generalized CADiS designed as in Fig.1. To ensure the connectivity between  $\mathcal{G}_1$  and its symmetric coarray, the GHOST is to add the third subarray  $\mathcal{P}_3$ , whose positions yield

$$\mathcal{P}_3 = \{\beta - \alpha, \beta m, m \in \langle 2, Q_0 \rangle\}, \quad (11)$$

where  $Q_0 = \max\{\lceil(\alpha - 1)/2\rceil, 2\}$ ,  $\beta > \alpha$ . When  $\alpha \in \langle 1, M + 1 \rangle$  and  $\beta \in \langle 1, N \rangle$ , the closed-form expressions for the range of uDOFs and the number of DOFs are denoted as  $\mathcal{U}_c^G$  and  $U_d^G$ , yielding

$$\mathcal{U}_c^G = \langle -U_2^G + U_0, U_2^G - U_0 \rangle, \quad (12)$$

$$U_d^G = 2U_2^G - U_0 - 1, \quad (13)$$

where  $U_2^G = \alpha(N + 1) + \beta M$  and  $U_0 = (\alpha - 1)(\beta - 1)$ .

Theorem 1 can be proved similar to the hole-filling part in [22], so we omit the proof here.

Note that the GHOST can fill the probable holes in  $\mathcal{G}_1$  within  $\langle 0, U_0 \rangle$ . However, there still exist some holes within  $\langle U_2^G - U_0, U_2^G \rangle$  and we need to further add the fourth subarray after the second one to fill the holes and extend the virtual aperture. Similarly to the design of  $\mathcal{P}_3$ , the positions of the fourth subarray are defined as

$$\mathcal{P}_4 = \{\max(\mathcal{P}_2) + \alpha P_1 + \beta m, m \in \langle 1, Q_1 \rangle\}, \quad (14)$$

where  $P_1 \in \mathbb{Z}^+$  and  $Q_1$  denotes the number of sensors in the fourth subarray. As the introduction of  $\mathcal{P}_4$  may generate some other holes, it is necessary to implement the augmentation of additional sensors. Here, we define the new sparse array as the improved GHOST, namely GHOST-1, in the following theorem.

*Theorem 2:* Suppose the fourth subarray given as (14). Intending to form the increased uDOFs, it is necessary to add  $P_1$  sensors with positions

$$\mathcal{P}_5 = \{\max(\mathcal{P}_4) + \alpha + (\beta - \alpha)m, m \in \langle 1, P_1 \rangle, P_1 \geq 1\}. \quad (15)$$

When  $\alpha \in \langle 1, M + 1 \rangle, \beta \in \langle 1, N \rangle$ , GHOST-1 can construct the enlarged coarray with the range of uDOFs  $\mathcal{U}_c^{G1}$  and the number of DOFs  $U_d^{G1}$ , which satisfy

$$\mathcal{U}_c^{G1} = \langle -U_2^{G1} + U_0, U_2^{G1} - U_0 \rangle, \quad (16)$$

$$U_d^{G1} = 2U_2^{G1} - U_0 - P_1(P_1 + 1) + 1, \quad (17)$$

where  $U_2^{G1} = \alpha(N + P_1 + 1) + \beta(M + Q_1 + P_1)$ . The feasible domain of  $Q_1$  is provided in the following proof.

*Proof:* We define the DCAs between  $\mathcal{P}_i$  and  $\mathcal{P}_j$  ( $i \geq j$ ) as  $\mathcal{G}_{ij}$ . Using all the DCAs, we can formulate the parameter distribution of GHOST-1 with some missing lags in  $\{\mathcal{H}_i\}_{i=1}^4$ , which restrict the construction of a larger generic-coarray and the extension of the virtual aperture. The missing areas can be represented as

$$\begin{aligned} \mathcal{H}_1 &= \{h_1 \mid h_1 = \alpha n + \beta m, m \in \langle M + 1, Q_1 + P_1 - n \rangle \\ &\quad n \in \langle 1, \min(Q_1 + P_1 - M - 1, P_1 - 1) \rangle\}, \end{aligned}$$

$$\mathcal{H}_2 = \{h_2 \mid h_2 = \alpha n + \beta(M + Q_1 + P_1 + 1 - n), n \in \langle 1, P_1 \rangle\},$$

$$\mathcal{H}_3 = \{h_3 \mid h_3 = \alpha(N + 1 + n) + \beta m, n \in \langle 1, P_1 \rangle,$$

$$m \in \langle M + Q_1 + P_1 + 1 - n, M + Q_1 + P_1 \rangle\},$$

$$\mathcal{H}_4 = \{h_4 \mid h_4 = \alpha n + \beta M, n \in \langle N + 1, N + P_1 - 1 \rangle\}.$$

Therefore, it is necessary to demonstrate that all holes can either be filled by existing virtual lags or lie outside the consecutive region of the enlarged generic-coarray.

If  $P_1 = 1$ , there are no holes in  $\mathcal{H}_1$  or  $\mathcal{H}_4$ . All the regions can be connected except at  $\alpha + \beta(M + Q_1 + 1)$  and  $\alpha(N + 2) + \beta(M + Q_1 + 1)$ , where holes are present. Denoting  $\mathcal{G}_4$  as

$$\mathcal{G}_4 = \{g_4 \mid g_4 = \alpha n + \beta m, n \in \langle 0, N + 1 \rangle, m \in \langle 0, M + Q_1 \rangle\},$$

we can prove  $\mathcal{G}_4$  having all lags in  $\langle U_0, U_2^{g_4} - U_0 \rangle$ , where  $U_2^{g_4} = \alpha(N + 1) + \beta(M + Q_1)$ . We find  $\alpha + \beta(M + Q_1 + 1) \in \langle U_0, U_2^{g_4} - U_0 \rangle$ , that is  $\alpha + \beta(M + Q_1 + 1)$  can be filled by existing lags. Then we can define the coarray  $\mathcal{G}_5$  as

$$\mathcal{G}_5 = \{g_5 \mid g_5 = \alpha n + \beta m, n \in \langle 0, N + 2 \rangle, m \in \langle M, M + Q_1 + 1 \rangle \setminus \{\alpha(N + 2) + \beta(M + Q_1 + 1)\}\}. \quad (18)$$

which has consecutive lags within  $\langle U_1^{g_5} + U_0, U_2^{g_5} - U_0 \rangle$ ,  $U_1^{g_5} = \beta M$ ,  $U_2^{g_5} = \alpha(N + 2) + \beta(M + Q_1 + 1) = U_2^{G_1}$  with  $\alpha \leq Q_1 + 2$ . Since  $U_2^{g_4} - U_0 > U_1^{g_5} + U_0$ , it holds true for the results in (16) when  $Q_1 \geq \alpha - 2$ .

If  $P_1 \in [2, \alpha - 1]$ , considering  $\mathcal{H}_1$ , if  $Q_1 + P_1 \leq M + 1$ , there are no holes to be filled. If  $Q_1 + P_1 > M + 1$ , we recursively define the coarray  $\mathcal{G}_5^i$  as

$$\mathcal{G}_5^i = \{g_5^i \mid g_5^i = \alpha n + \beta m, n \in \langle 0, N \rangle, m \in \langle 0, i \rangle\}, \\ i \in \langle M, Q_1 + P_1 - 2 \rangle,$$

and discuss the following three cases.

- Case 1 If  $P_1 - 1 + \beta \leq N$ , it can be proved that  $\mathcal{G}_5^i$  has all lags in  $\langle U_0, U_2^{g_5^i} - U_0 \rangle$ , where  $U_2^{g_5^i} = \alpha N + \beta i$ . Since  $\alpha(P_1 - 1) + \beta(i + 1) \leq U_2^{g_5^i} - U_0$  and  $\alpha + \beta(i + 1) \geq U_0$  always holds when  $P_1 - 1 + \beta \leq N$ , all holes in  $\mathcal{H}_1$  with  $m = i + 1$  can be recursively filled.
- Case 2 If  $P_1 - 1 + \beta > N$ ,  $Q_1 \leq M$ , when  $Q_1 + P_1 + \beta \leq M + N + 2$ , we can find that  $\alpha(Q_1 + P_1 - M - 2) + \beta(i + 1) \leq U_2^{g_5^i} - U_0$  and  $\alpha + \beta(i + 1) \geq U_0$ . Besides, in this case  $\alpha(Q_1 + P_1 - M - 1) + \beta(M + 1) \leq \alpha(N + 1) + \beta M + \alpha - U_0$ , then all holes in  $\mathcal{H}_1$  can be filled.
- Case 3 If  $P_1 - 1 + \beta = N + 1$ ,  $Q_1 \geq M + 1$ ,  $\alpha(P_1 - 2) + \beta(i + 1) \leq U_2^{g_5^i} - U_0$  always holds. When  $Q_1 \leq M + \alpha - 2$ ,  $\alpha(P_1 - 1) + \beta(Q_1 + 1) \leq \alpha(N + 1) + \beta M - U_0$  and all holes in  $\mathcal{H}_1$  are filled.

Considering  $\mathcal{H}_2$ , we can discuss the following three cases.

- Case 1 If  $P_1 + \beta \leq N$ , the holes in  $\mathcal{H}_2^i = \{\alpha i + \beta(M + Q_1 + P_1 + 1 - i)\}_{i=1}^{P_1}$  can be recursively filled by  $\mathcal{G}_5^{s_i} = \{g_5^{s_i} \mid g_5^{s_i} = \alpha n + \beta m, n \in \langle 0, N \rangle, m \in \langle 0, M + Q_1 + P_1 - i \rangle\}$ .
- Case 2 If  $P_1 + \beta = N + 1$ ,  $Q_1 \neq (\alpha - 1)$ , the holes in  $\mathcal{H}_2^i$  can be recursively filled by  $\mathcal{G}_5^{c_i} = \{g_5^{c_i} \mid g_5^{c_i} = \alpha n + \beta m, n \in \langle 0, N + 1 \rangle, m \in \langle 0, M + Q_1 + P_1 - i \rangle\}$ .
- Case 3 If  $P_1 + \beta > N + 1$ ,  $\beta < N$ , the holes in  $\mathcal{H}_2^i$  can be recursively filled by  $\mathcal{G}_5^{e_i} = \{g_5^{e_i} \mid g_5^{e_i} = \alpha n + \beta m, n \in \langle i, N + i \rangle, m \in \langle M + Q_1, M + Q_1 + P_1 - i \rangle\}$  with  $Q_1 \geq \alpha$ , while if  $\beta = N$ , they can be filled with  $Q_1 \geq \alpha - 1$ .

With holes in  $\mathcal{H}_1$  and  $\mathcal{H}_2$  filled, we define the coarray  $\mathcal{G}_5^h$  as

$$\mathcal{G}_5^h = \{g_5^h \mid g_5^h = \alpha n + \beta m, n \in \langle 0, N + P_1 + 1 \rangle, m \in \langle M + 1, M + Q_1 + P_1 \rangle\},$$

which has consecutive lags in  $\langle U_1^{g_5^h} + U_0, U_2^{g_5^h} - U_0 \rangle$ , where  $U_2^{g_5^h} = \alpha(N + P_1 + 1) + \beta(M + Q_1 + P_1)$ . It can be proved that the smallest value of holes in  $\mathcal{H}_3$ , i.e.  $\min(\mathcal{H}_3) = \alpha(N + P_1 + 1) + \beta(M + Q_1 + 1) > U_2^{g_5^h} - U_0$  when  $P_1 \leq \alpha - 1$ , which means that all holes in  $\mathcal{H}_3$  can lie outside the consecutive region of  $\mathcal{G}_5^h$ .

When  $\alpha \leq Q_1 + P_1$ , as  $\min(\mathcal{H}_4) = \alpha(N + 1) + \beta M \geq U_1^{g_5^h} + U_0$  and  $\max(\mathcal{H}_4) = \alpha(N + P_1 - 1) + \beta M \leq U_2^{g_5^h} - U_0$ , all holes in  $\mathcal{H}_4$  are filled by  $\mathcal{G}_5^h$ .

Finally, all coarrays can be connected and the results in Theorem 2 hold true when all conditions before are satisfied. In this case, the range of  $Q_1$  can be summarized as follows:

$$\begin{cases} Q_1 \in \mathcal{Q}_1, & \beta = N \\ Q_1 \in \mathcal{Q}_2, & \beta \in \langle N - P_1 + 2, N - 1 \rangle \\ Q_1 \in \mathcal{Q}_3, & \beta = N - P_1 + 1 \\ Q_1 \geq \alpha - P_1, & \beta \in \langle \alpha + 1, N - P_1 \rangle \end{cases}, \quad (19)$$

where  $P_1 \in \langle 2, \alpha - 1 \rangle$ ,  $\mathcal{Q}_1 = \langle \alpha - 1, M + 2 - P_1 \rangle$ ,  $\mathcal{Q}_2 = \langle \alpha, M + N + 2 - P_1 - \beta \rangle$ ,  $\mathcal{Q}_3 = \langle \alpha - P_1, \alpha - 2 \rangle \cup \langle \alpha, \infty \rangle$ . Besides, when  $\beta = N - P_1 + 2$ , the feasible domain of  $Q_1$  additionally includes the interval  $\mathcal{Q}_4 = \langle M + 1, M + \alpha - 2 \rangle$ . ■

#### IV. OPTIMIZATION ALGORITHM

It is shown that GHOST-1 can obtain more virtual lags by adding another  $(Q_1 + P_1)$  sensors than GHOST, which enables us to explore the maximum uDOFs for a given sensor number  $T$ , bringing about the optimization problem as follows:

$$\begin{aligned} \max_{\alpha, \beta, P_1} \quad & U_2^{G_1} - U_0 + 1 \\ \text{s.t.} \quad & \alpha \in \langle 1, M + 1 \rangle, \beta \in \langle \alpha + 1, N \rangle, P_1 \in \langle 2, \alpha - 1 \rangle \\ & T = M + N + Q_0 + Q_1 + P_1. \end{aligned} \quad (20)$$

The solution of (20) provides a GHOST-1 configuration with desired sensor number  $T$ . According to (17), sensors in  $\mathcal{P}_5$  contribute the most to the increase in uDOFs, thus we can set  $Q_1$  to be the minimum value of its feasible domain in (19). Subsequently, the optimal solution of (20) is derived in the following corollary.

*Corollary 1:* Suppose the GHOST-1 in Theorem 2 designed with  $P_1 \in \langle 2, \alpha - 1 \rangle$ . Then, for a given sensor number  $T$  and fixed parameter  $\alpha$ , the maximum uDOFs is obtained when  $\beta = N$  as

$$\sup \mathcal{U}_c^{G_1} = -N^2 + (T - Q_0 + 1)N + \alpha(P_1 + 2) - 1 \quad (21)$$

*Proof:* With  $Q_1$  set as the minimum value of its feasible domain in (19), we can discuss the following three cases.

Case 1 Suppose  $\beta = N$ , the objective function will be

$$\begin{aligned} f(\alpha, \beta) &= -(N - N_0)^2 + N_0^2 + \alpha(P_1 + 2) \\ &\leq N_0^2 + \alpha(P_1 + 2) \end{aligned}$$

where the optimal solution is at  $N = N_0 = (T - Q_0 + 1)/2$ . As  $\mathcal{Q}_1$  in (19) is not an empty set, it holds that  $P_1 \leq M + 3 - \alpha$ , and the optimal solution is  $P_1 = \min\{M + 3 - \alpha, \alpha - 1\}$ .

Case 2 Suppose  $\beta = N - x$ ,  $x \in \langle 1, P_1 - 2 \rangle$ , the objective function is

$$\begin{aligned} f(\alpha, \beta) &= -(N - \frac{T - Q_0 + 1}{2})^2 + (\frac{T - Q_0 + 1}{2})^2 \\ &\quad + \alpha(P_1 + 2) - x(M + Q_1 + P_1 - \alpha + 1) \\ &\leq N_0^2 + \alpha(P_1 + 2) - x(M + Q_1 + P_1 - \alpha + 1) \end{aligned}$$

where the optimal solution is at  $N = N_0 = (T - Q_0 + 1)/2$ . As  $f(\alpha, \beta)$  monotonically decreases with  $x$ , it holds that

$$f(\alpha, \beta) \leq N_0^2 + \alpha P_1 + 2\alpha - (x_0 - \alpha + 1)$$

where  $x_0 = M + P_1 + Q_1$  is a determinate value. Then the optimal solution is obtained at the maximum  $P_1$ , i.e.,  $P_1 = \min\{\alpha - 1, T - N - Q_0 - 2\alpha + 1\}$  when  $P_1 \geq 3$ . The uDOFs in this case is smaller than that in Case 1.

Case 3 Suppose  $\beta = N - x$ ,  $x \in \langle P_1 - 1, N - \alpha - 1 \rangle$ , similarly we can get the objective function as

$$\begin{aligned} f(\alpha, \beta) &\leq N_0^2 + \alpha(P_1 + 2) \\ &\quad - (P_1 - 1)(M + Q_1 + P_1 - \alpha + 1) \\ &= N_0^2 + (2\alpha - x_0 - 1)P_1 + \alpha + x_0 + 1 \end{aligned} \quad (22)$$

where the optimal solution is at  $N = N_0 = (T - Q_0 + 1)/2$ .  $x_0 = M + P_1 + Q_1 \geq 2\alpha - 1$  is a determinate value and the function monotonically decreases with  $P_1$ . Then the optimal solution is at  $P_1 = 2$ ,  $Q_1 = \alpha - P_1$ . The uDOFs in this case can also be proved to be smaller than that in Case 1.

Above all,  $\beta = N$  is the optimal solution to get the largest uDOFs. ■

With Corollary 1, we can get the optimal solution as

$$\begin{aligned} \beta &= N \rightarrow \frac{(T - Q_0 + 1)}{2}, Q_1 = \alpha - 1, \\ P_1 &\leq \min\{\alpha - 1, M + 3 - \alpha\}, \\ M &= T - N - Q_0 - Q_1 - P_1 \geq \alpha - 1. \end{aligned} \quad (23)$$

and  $\beta$  is a prime number with  $\alpha$ .

## V. NUMERICAL SIMULATIONS

In this section, we compare the proposed array configured via GHOST-1 with EGCA [22] in terms of uDOFs and MC. As the prototype array of GHOST-1 in this paper is a generalized CADiS which is similar to EGCA, EGCA can be seen as a special case of GHOST-1 with  $Q_1 = M$ ,  $P_1 = 1$  and without  $P_5$ . The uDOF ratio  $\gamma(T)$  can be defined as  $\gamma(T) = T^2/L(T)$ , where  $L(T)$  is the maximum number of uDOFs in the positive part of the DCA. The smaller  $\gamma(T)$  is, the higher capacity can be obtained.

Fig. 2 illustrates the uDOF ratio and MC leakage of the proposed array and EGCA versus  $T$  with  $c = 0.3$  and  $B = 100$  under different  $\alpha$ . It can be seen that, On the whole, a larger value of  $\alpha$  leads to a reduction in MC leakage at the expense of a decrease in uDOFs. One potential problem in Fig. 2 is that the uDOF ratio of GHOST-1 with  $\alpha = 7$  exhibits a valley even lower than that of  $\alpha = 5$ , which is attributed to the fact that a larger  $\alpha$  is more conducive to aperture expansion, and with a moderate number of sensors, fewer additional sensors are required to fill holes, resulting in less redundancy. Due to the similarity between EGCA and the proposed array, their MC leakage is nearly identical for the same  $\alpha$ . However, the proposed array can achieve higher uDOF capacity, especially when the number of sensors  $T$  is small.

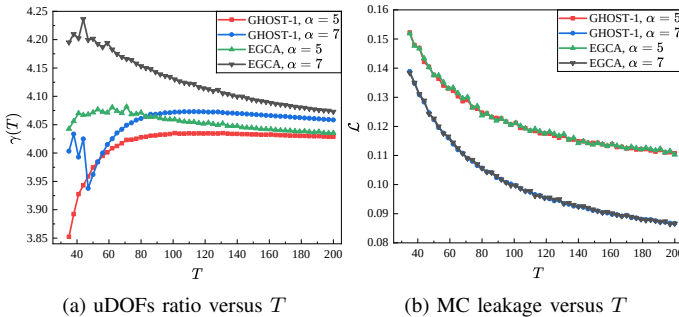
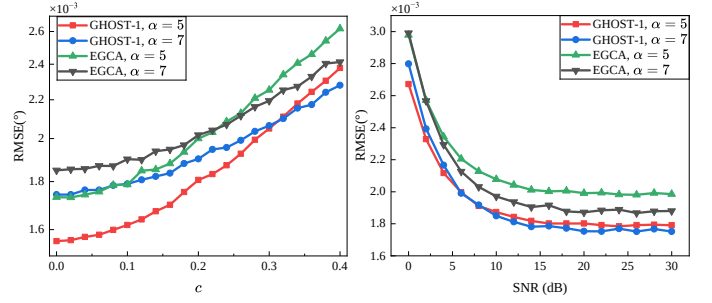


Fig. 2. uDOFs ratio and MC leakage versus sensor number  $T$  with  $c = 0.3$  and the number of snapshots  $L = 1000$ .

Then, we investigate the DOA estimation accuracy under MC effect via 1000 Monte Carlo trials. All sparse arrays are configured

with  $T = 35$  sensors and  $K = 30$  uncorrelated sources are evenly distributed within  $[-60^\circ, 60^\circ]$ . Fig. 3(a) depicts the root mean square error (RMSE) of estimation versus the MC strength  $c$ , with the signal-to-noise ratio (SNR) as 10dB and the number of snapshots as  $L = 1000$ . Due to the higher uDOFs, GHOST-1 always outperforms EGCA with the same  $\alpha$ . For small  $c$ , estimation accuracy is dominated by uDOFs, leading to better performance of arrays configured with  $\alpha = 5$ . However, as  $c$  increases, the accuracy of arrays with  $\alpha = 7$  degrades more slowly and GHOST-1 with  $\alpha = 7$  even outperforms all the other arrays when  $c > 0.3$ . Fig. 3(b) shows the RMSE versus SNR with  $c = 0.3$  and  $L = 1000$ , and it can be seen that the proposed arrays achieve higher accuracy under different SNR conditions.



(a) RMSE versus  $c$  with SNR = 10dB (b) RMSE versus SNR with  $c = 0.3$

Fig. 3. RMSE performance in different scenarios.

## VI. CONCLUSIONS

In this paper, we propose a novel hole-filling strategy for sparse coprime-type arrays. The approach employs augmented subarrays to eliminate inherent holes caused by parameter coprimality through the construction of an enlarged generic-coarray, thus extending the virtual aperture and achieving enhanced uDOFs. Simulation results demonstrate the superiority of sparse arrays configured with the proposed strategy in terms of uDOFs and DOA estimation accuracy.

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